

B. Tech. (3rd Sem)

(Common for CSE, CSE with Specialization in Data Science, CSE with specialization in Cloud Technology & Information Security, CSE with specialization in Big Data and Analytics, CSE with specialization in Full Stack Development, CSE with specialization in AI & Machine Learning)

BCSE-502I (Software Engineering)

L	T	P	Continuous evaluation	40
3	0	0	End semester exam	60
			Total marks	100
			Credits	3.0

Course Objective:

1. To impart necessary and practical knowledge of software engineering and testing and develop skills required as a software tester to test real-life projects.
2. To provide an idea of using various process models in the software industry.
3. Be successful professionals in the field with solid fundamental knowledge of software engineering and testing
4. To study fundamental concepts in software testing, including software testing objectives, process, criteria, strategies, and methods.
5. To discuss various software testing issues and solutions in software unit test; integration, regression, and system testing.
6. To learn how to plan a test project, design test cases and data, conduct testing operations, manage software problems and defects, generate a testing report.
7. To understand software test automation problems and solutions.
8. To learn how to write software testing documents, and communicate with engineers in various forms.
9. To gain the techniques and skills on how to use modern software testing tools to support software testing projects.

Unit:-1

(9 Hours)

Introduction to software and software engineering, Software characteristics, Software engineering paradigms (SDLC Models Waterfall model, Incremental Model, Spiral, Prototype, Big Bang, V model, Agile Model-Devops), Comparison of all SDLC models, Software failure & crisis.

Planning a Software project: Software cost estimation, Project scheduling, Personal planning, Team structure, Software configuration management, quality assurance, project monitoring, risk management.

Unit:-2

(9 Hours)

Software Requirement Analysis: Structured Analysis, Object Oriented Analysis, Software requirement specification, Validation.

Design and Implementation of Software: Software design fundamentals, design methodology (structured design and object oriented design), Design verification.

Introduction to UML: USE Case Diagrams, Object Diagrams, Class Diagrams.

Unit:-3

(9 Hours)

Software Testing - Importance & As a Career: Testing Fundamentals, Software Testing Life Cycle (STLC), Aligning testing with project, Cyclomatic complexity. Software testing types: Static testing (Reviews, Inspection, Walkthroughs), Dynamic Testing, White box testing, Black box testing, Risk based testing.

Software Testing Strategies: Unit testing, Integration testing, Validation testing, System testing, Performance testing, Regression testing, Stress testing, Smoke testing, Mutation testing, Regular expression based testing, Debugging.

Unit:-4

(9 Hours)

Test Phases, Test Scenarios, Requirement Traceability Matrix (RTM), ETVX Model (Entry, Task, Verification and Exit model). Test cases: Introduction, creating test cases using white box, black box techniques. Flow graphs & Transaction flow. Test Plan and estimation Techniques: Introduction to Test Plan, Test Environment and Schedule, Staffing and test deliverables, Software test estimation Techniques. Test oracle, Concurrency & Transaction Analysis.

Quality management, CMMi, TMMi Model and comparison. **Test Automation:** Need for Automation, Categorization of testing tools, Selection of testing tools.

Testing Tools: Introduction to various testing tools: Junit, Jira, Mantis, Bugzilla, Test complete, and Generation of test cases and simulation of test cases.

Course Outcome: After the completion of this course, the students will be able to:

1. Gather and specify requirements of the software projects.
2. Design software requirements with existing tools.
3. Identify complex engineering problems by applying principles of software engineering and testing.
4. Create various documents, and deliverables related to software testing.
5. Differentiate between different types of testing.
6. Acquire and apply new knowledge as needed, using appropriate learning strategies.
7. Work as an individual and as part of a multidisciplinary team to develop and deliver quality software.

Instructions for paper setter: All Questions are compulsory. The Question paper is divided in to four sections A, B, C and D. Section A is compulsory and comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. Section B comprises of 4 questions of 2 marks each, one from each unit. Section C Comprises of 4 questions of 4 marks each, one from each unit. Section D Comprises of 4 questions of 6 marks each, one from each unit. There is no overall choice, however internal choice may be provided in section C and D, if paper setter so desires.

Text Books:

1. Software Engineering by Pressman S. Roger, Tata McGraw-Hill.
2. Software Engineering by K.K Aggarwal, New Age.
3. An integrated approach to software engineering by Jalote Pankaj, Narosa Publishing House.
4. Software Testing: Principles and Practices by Srinivasan Desikan.
5. Software Automation Testing Tools for Beginners by Rahul Shende.
6. Foundations of Software Testing ISTQB Certification by Rex Black, Erik van Veenendaal, Dorothy Graham.

Reference Books:

1. Software, Software Engineering Concepts by Fairley Richard, Tata McGraw-Hill.
2. Software Engineering by Somerville Ian, 5th ed., Addison Wesley-2000.
3. The Essentials of Modern Software Engineering: Free the Practices from the Method Prisons! (ACM Books) by Ivar Jacobson, Harold "Bud" Lawson, Pan-Wei Ng.
4. Automated Software Testing: Foundations, Applications and Challenges (Services and Business Process Reengineering) by Ajay Kumar Jena, Himansu Das, Durga Prasad Mohapatra.
5. Software Quality and Testing: A Concise Study by Kelkar S.A.
6. Foundations of Software Testing by Dorothy Graham, Isabel Evans.